

WASP-43b

R-1771

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220218 · created 2026-08-02T22:07:51+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459628.7817 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.015
Transit depth (Rp/Rs)^2	3.93 +/- 0.58 [%]
Orbital Inclination (inc)	82.04 +/- 1.42
Airmass coefficient 1 (a1)	1.011 +/- 0.029
Airmass coefficient 2 (a2)	-0.009 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.91 %
Best Comparison Star	#3 - [247, 100]
Optimal Method	PSF photometry
Transit Duration (day)	0.0527 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.198 ± 0.015 vs 0.1594 (deviation 1.59σ)
Duration fit vs published	0.0527 d vs 0.0512 d (deviation 0.2σ)
Mid-time O-C	5.3 min
Depth SNR	8.1
Residual scatter	2.91 %

Light curve

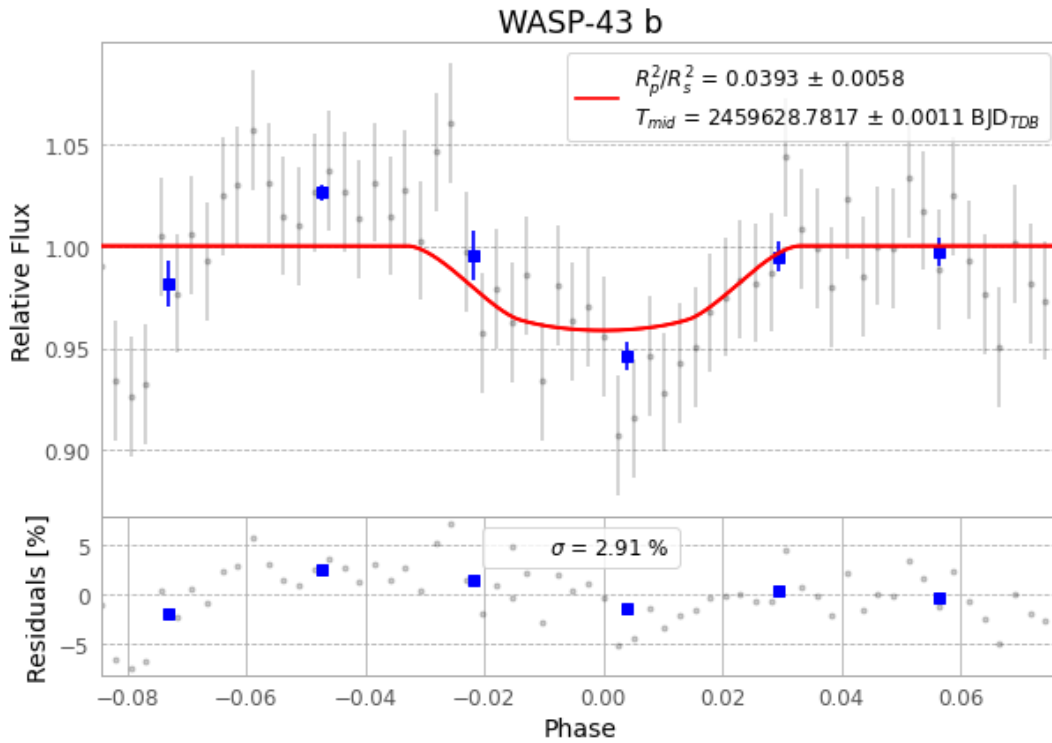


Figure 1 — FinalLightCurve_WASP-43 b_2022-02-17.png (SHA-256 84ee2f9a34254cf1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 295], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ed0c596d53562ef5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43220218045712.FITS → WASP-43220218080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220218.log (c0a9a2828e2b...), FinalLightCurve_WASP-43 b_2022-02-17.png (84ee2f9a3425...), FinalLightCurve_WASP-43 b_2022-02-17.csv (925842b952fc...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 22:07Z.